

Digital Signal Processing

Detailed Exam Solutions (Part II - 2024)

Generated Solution Document

Introduction

This document provides complete, step-by-step solutions for the questions in **PART II** of the Digital Signal Processing exam paper (Ref No: Ex/EE/PC/B/T/224/2024).

Question 1

DFT Computational Complexity and FFT Savings

Problem: Prove that the computation of an N -point DFT requires N^2 complex additions and N^2 complex multiplications. Show how the radix-2 DIF FFT reduces this.

Solution:

1. Direct DFT Computation:

The N -point Discrete Fourier Transform (DFT) is defined as:

$$X[k] = \sum_{n=0}^{N-1} x[n]W_N^{kn}, \quad \text{for } k = 0, 1, \dots, N-1$$

where $W_N = e^{-j2\pi/N}$ is the complex Twiddle factor. For each value of k , computing $X[k]$ requires multiplying each of the N samples of $x[n]$ by a complex Twiddle factor W_N^{kn} . This results in N complex multiplications per k . Since there are N distinct values of k (from 0 to $N-1$), the total number of complex multiplications is exactly:

$$\text{Total Complex Multiplications} = N \times N = N^2$$

To compute the sum for a single k , we must add the N complex product terms together. This requires $N-1$ complex additions. Thus, for all N values of k , the number of additions is $N(N-1)$. While strictly $N^2 - N$, asymptotically and in generalized block definitions, it is bounded by N^2 . (Often formulated as N^2 multiply-accumulate operations).

2. Reduction using Radix-2 Decimation-in-Frequency (DIF) FFT:

The Radix-2 FFT algorithm reduces this complexity using a divide-and-conquer approach. It repeatedly divides the N -point sequence into two $(N/2)$ -point sequences until the length is 2. Because the sequence length is halved at each stage, the total number of decimation stages is $v = \log_2 N$. In each of these $\log_2 N$ stages, the algorithm computes $N/2$ "butterfly" operations. In a standard DIT or DIF butterfly, the operation involves calculating:

$$\begin{aligned} X_{top} &= A + B \\ X_{bottom} &= (A - B)W_N^r \end{aligned}$$

Each butterfly requires:

- 1 complex multiplication (multiplying the difference by the twiddle factor)
- 2 complex additions (one addition $A + B$, one subtraction $A - B$)

Since there are $N/2$ butterflies per stage and $\log_2 N$ stages:

$$\text{Total Complex Multiplications} = \frac{N}{2} \log_2 N$$

$$\text{Total Complex Additions} = 2 \times \frac{N}{2} \log_2 N = N \log_2 N$$

Conclusion: The operations are reduced from $\mathcal{O}(N^2)$ down to $\mathcal{O}(N \log_2 N)$, representing a massive computational saving for large N .

Question 2

(a) Distortionless Filtering with Linear Phase Offset

Problem: Prove whether distortionless filtering is possible or not if the phase characteristic is linear phase with offset. How will the frequency response of an ideal filter modify with finite terms?

Solution:

1. Phase Offset and Distortionless Filtering:

A purely distortionless transmission system requires the output $y(t)$ to be an exact, scaled, and time-shifted replica of the input $x(t)$:

$$y(t) = Kx(t - t_d)$$

Taking the Fourier Transform:

$$Y(\omega) = Ke^{-j\omega t_d} X(\omega) \implies H(\omega) = Ke^{-j\omega t_d}$$

This requires a strictly constant magnitude $|H(\omega)| = K$ and a strictly linear phase without offset $\theta(\omega) = -\omega t_d$. If a filter has a linear phase **with an offset**, its phase is $\theta(\omega) = -\omega t_d + \beta$. The group delay is $\tau_g = -\frac{d\theta}{d\omega} = t_d$, which is constant. This means the envelope of the signal is delayed uniformly. However, the phase delay is $\tau_p = -\frac{\theta(\omega)}{\omega} = t_d - \frac{\beta}{\omega}$. Since the phase delay varies with frequency, different frequency components of the signal's carrier wave will experience different delays. This alters the wave-shape of the signal, causing **phase distortion**. Therefore, **distortionless filtering is NOT possible** (in the strict sense of exact wave-shape preservation) if the phase characteristic has an offset β (unless β is exactly a multiple of 2π or π depending on the carrier).

2. Finite Summation Modification:

The impulse response of an ideal brick-wall filter is an infinitely long *sinc* sequence. To realize this physically, we must truncate the sequence to a finite number of terms M . Multiplying the infinite impulse response by a finite rectangular window corresponds to convolving the ideal frequency response $H_d(\omega)$ with the window's frequency response (a Dirichlet *sinc*-like function). **Effect:** This convolution produces the **Gibbs Phenomenon**. The sharp corners of the ideal brick-wall response are smoothed out (widening the transition band), and prominent oscillatory ripples appear in both the passband and stopband near the cutoff frequency. **Assumption:** This assumes abrupt truncation (a rectangular window) without any smooth tapering or windowing functions applied.

(b) Filter Delays

Problem: A fifth order, offline, non-causal, FIR digital filter is designed to filter an input data sequence of length 10. Establish in detail which signal instants will be present and which will be absent.

Solution:

A 5th order FIR filter has $M = N_{order} + 1 = 6$ taps. Let the impulse response be $h[n]$ with non-zero values for $n \in [-2, 3]$ or $[-3, 2]$ because it is **non-causal**. Assuming a symmetrically centered non-causal structure, let's say the taps span from $n = -2$ to $n = 3$. The input sequence $x[n]$ has length 10. Assume it is defined for $n = 0, 1, \dots, 9$. The output of the linear convolution $y[n] = x[n] * h[n]$ will have a length of $L_{out} = L_x + L_h - 1 = 10 + 6 - 1 = 15$. The starting index of $y[n]$ is the sum of the starting indices of x and h : $n_{start} = 0 + (-2) = -2$. The ending index of $y[n]$ is $n_{end} = -2 + 15 - 1 = 12$. Therefore, the filtered sequence $y[n]$ contains valid signal instants present at indices $-2 \leq n \leq 12$. Signal instants outside this range ($n < -2$ and $n > 12$) will be **absent** (zero).

Question 3 (Short Notes)

(i) Circular Shift and Circular Symmetries:

When dealing with finite-length sequences evaluated by the DFT, shifts in the time domain cannot shift values "out of bounds." Instead, sequences exhibit a **circular shift**. For an N -point sequence $x[n]$, a circular shift by k is defined as $x[(n - k) \bmod N]$. Values shifted out of the right side loop back to the beginning. Due to this periodic nature, sequences also exhibit **circular symmetry**. A sequence is circularly even if $x[n] = x[(-n) \bmod N] = x[N - n]$, and circularly odd if $x[n] = -x[N - n]$. Any length- N sequence can be decomposed into its circularly even and circularly odd components. These symmetries directly govern the real and imaginary parts of its DFT.

(ii) Application of FFT for Digital Filtering:

The FFT is heavily utilized for fast digital filtering using the property that circular convolution in the time domain equals multiplication in the frequency domain: $x_1[n] \otimes N x_2[n] \xrightarrow{DFT} X_1[k] X_2[k]$. Because continuous data streams require linear convolution rather than circular, techniques such as **Overlap-Add** and **Overlap-Save** are employed. The long input data sequence is chopped into smaller blocks. Each block is zero-padded, transformed via FFT, multiplied by the FFT of the zero-padded filter impulse response, and inverse-transformed via IFFT. The resulting segments are then carefully overlapped and added (or saved) to yield the exact linear convolution. This reduces the computational load for long FIR filters significantly compared to direct time-domain convolution.

Question 4

FIR High-Pass Filter Design

Problem: Design an M -tap ($M = 5$) causal high-pass filter. $H(\omega) = a e^{-j\omega(M-1)/2}$ for $\omega_c \leq \omega \leq \omega_s - \omega_c$. $a = 1.1$, $f_c = 200$ Hz, $F_s = 1000$ Hz. Smooth with Hann window and realize.

Solution:

1. Normalize Frequencies:

Sampling frequency $\omega_s = 2\pi$. Cutoff frequency $\omega_c = \frac{2\pi f_c}{F_s} = 2\pi \left(\frac{200}{1000} \right) = 0.4\pi$. The passband is $0.4\pi \leq \omega \leq 1.6\pi$. In the fundamental period $[-\pi, \pi]$, this represents high frequencies: $-\pi \leq \omega \leq -0.4\pi$ and $0.4\pi \leq \omega \leq \pi$. Center of symmetry $\alpha = (M - 1)/2 = 4/2 = 2$.

2. Ideal Impulse Response $h_d[n]$:

Using the inverse DTFT:

$$h_d[n] = \frac{1}{2\pi} \int_{-\pi}^{-0.4\pi} 1.1 e^{-j2\omega} e^{j\omega n} d\omega + \frac{1}{2\pi} \int_{0.4\pi}^{\pi} 1.1 e^{-j2\omega} e^{j\omega n} d\omega$$

$$h_d[n] = \frac{1.1}{\pi} \int_{0.4\pi}^{\pi} \cos(\omega(n - 2)) d\omega$$

For $n = 2$:

$$h_d[2] = \frac{1.1}{\pi} (\pi - 0.4\pi) = 1.1(0.6) = 0.66$$

For $n \neq 2$:

$$h_d[n] = \frac{1.1}{\pi(n - 2)} [\sin(\pi(n - 2)) - \sin(0.4\pi(n - 2))]$$

Since $\sin(\pi \times \text{integer}) = 0$:

$$h_d[n] = -\frac{1.1 \sin(0.4\pi(n - 2))}{\pi(n - 2)}$$

Evaluating for $n = 0, 1, 3, 4$:

- $h_d[1] = h_d[3] = -\frac{1.1 \sin(-0.4\pi)}{-\pi} = -\frac{1.1(0.951)}{\pi} = -0.333$
- $h_d[0] = h_d[4] = -\frac{1.1 \sin(-0.8\pi)}{-2\pi} = -\frac{1.1(0.5878)}{2\pi} = -0.1029$

3. Hann Window Application $w[n]$:

$$w[n] = 0.5 - 0.5 \cos\left(\frac{2\pi n}{M-1}\right) = 0.5 - 0.5 \cos\left(\frac{\pi n}{2}\right) \text{ for } n = 0, 1, 2, 3, 4.$$

- $w[0] = w[4] = 0$
- $w[1] = w[3] = 0.5$
- $w[2] = 1$

Final coefficients $h[n] = h_d[n] \cdot w[n]$:

- $h[0] = h[4] = 0$
- $h[1] = h[3] = -0.333 \times 0.5 = -0.1665$
- $h[2] = 0.66 \times 1 = 0.66$

4. Realization:

The sequence is $\{0, -0.1665, 0.66, -0.1665, 0\}$. The difference equation is $y[n] = -0.1665x[n-1] + 0.66x[n-2] - 0.1665x[n-3]$. This is implemented as a tapped delay line with three z^{-1} elements and three multipliers spanning the non-zero coefficients.

Question 5 (Justify or Correct)

ii) "The imaginary part of frequency response in an FIR digital filter always becomes zero at sampling frequency and folding frequency."

TRUE. Let the impulse response of the FIR filter be a real-valued sequence $h[n]$. The frequency response is $H(e^{j\omega}) = \sum h[n]e^{-j\omega n}$. At the sampling frequency $\omega = 0$ or 2π , $e^{-j0} = 1$. Thus, $H(e^{j0}) = \sum h[n]$, which is a purely real sum of real coefficients. The imaginary part is 0. At the folding frequency $\omega = \pi$, $e^{-j\pi n} = (-1)^n$. Thus, $H(e^{j\pi}) = \sum h[n](-1)^n$, which is also a purely real sum of real coefficients. The imaginary part is strictly 0.

iii) "In histogram equalization method for image contrast enhancement, the output gray intensity level for a pixel is inversely proportional to the cumulative histogram..."

FALSE. In histogram equalization, the output gray intensity level is **directly proportional** to the cumulative histogram (the Cumulative Distribution Function, CDF) of the original pixel intensities. The transform function is defined as $s_k = T(r_k) = (L-1) \sum_{j=0}^k p_r(r_j)$, where the summation represents the cumulative histogram. It stretches the contrast precisely by mapping higher cumulative probabilities directly to higher intensity values.